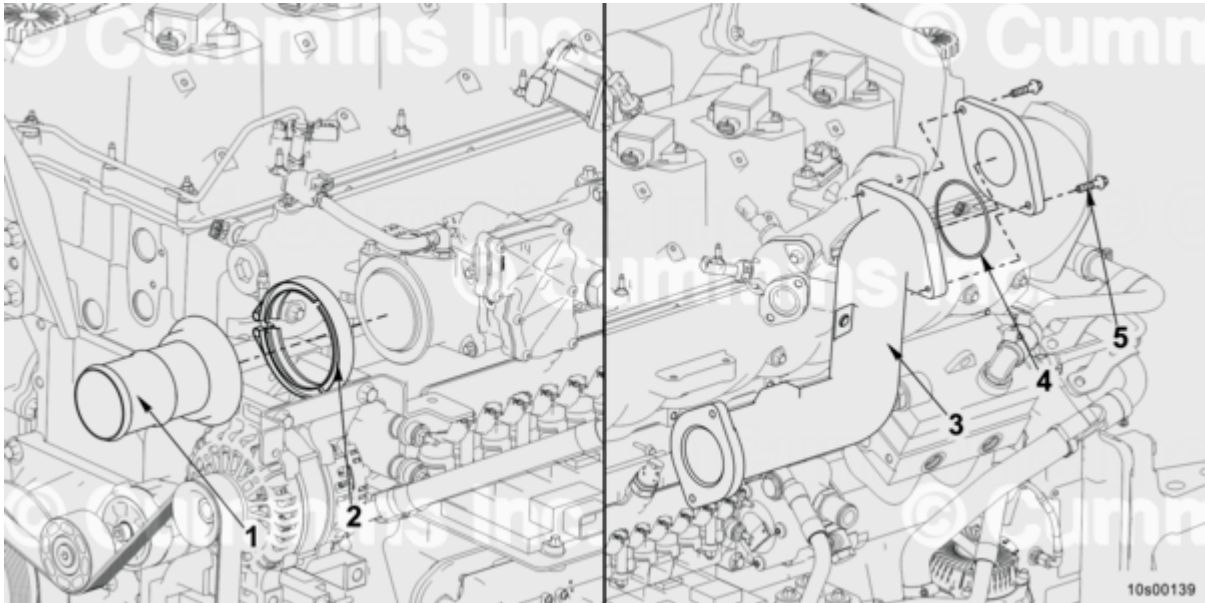


Trainings ▾

Exploded View



Air Intake Connection Exploded View

1. Air intake connection - quantity 1
2. V-band clamp - quantity 1
3. Air intake connection - quantity 1
4. O-ring - quantity 1
5. Capscrew - quantity 2

Select Service Tools

Recommended Cummins® Service Tools

- Air Handling Clean Care Kit, Part Number 4919588

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

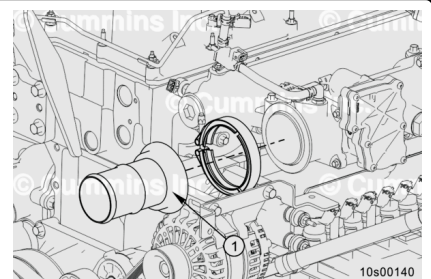
⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the charge air cooler connection. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/99/99-010-027.html>)
- Remove the engine intake manifold absolute pressure and temperature sensor. Refer to Procedure 019-099 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-099.html>)
- Remove the water jacket lines if equipped. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)

Remove

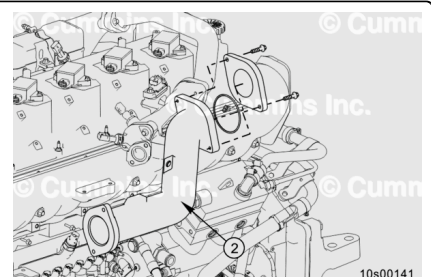
Remove the V-band clamp and the air intake connection (1) from the air intake connection adapter.



Remove the Exhaust Gas Recirculation (EGR) mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-021.html>)

Remove the two capscrews holding the air intake connection (2) to the air intake manifold.

Remove the air intake connection (2).

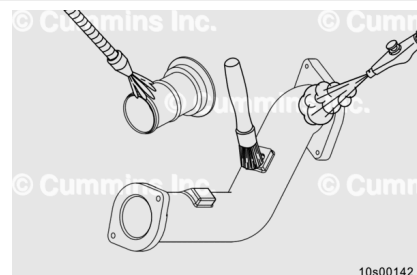


Use protective caps and heavy tape from the Air Handling Clean Care Kit, Part Number 4919588, to cover open points on the plumbing and engine.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



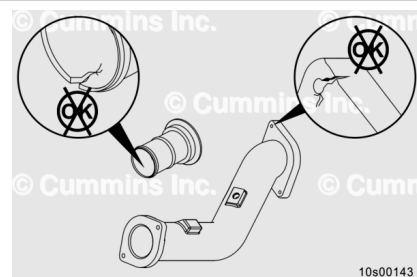
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

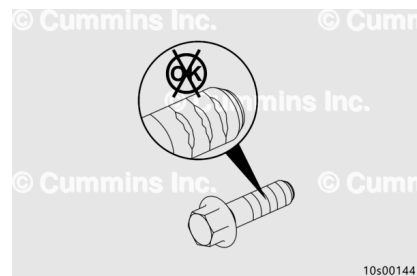
Clean the air intake connections (1) and (2). Saturate the inside of the air intake connections with solvent.

Dry the air intake connections with compressed air.

Inspect the air intake connections and clamp for cracks or other damage. Replace the connections and clamp if they are damaged.



Inspect the air intake connection capscrews for stripped threads or other damage. Replace the capscrews if they are damaged.

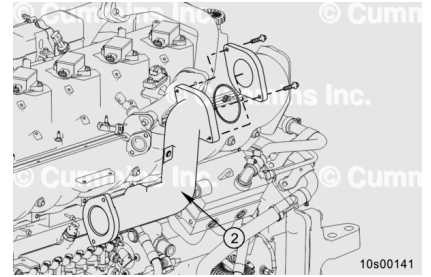


Install

Install the air intake connection (2) to the engine air intake manifold.

Tighten the two capscrews holding the air intake connection (2) to the air intake manifold.

Torque Value: 18 n•m [159 in-lb]

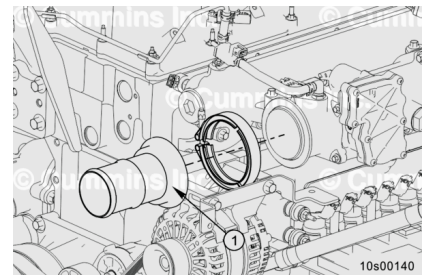


Install the EGR mixer. Refer to Procedure 011-021 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-021.html>)

Connect the air intake connection (1) to the air intake connection adapter.

Tighten the V-band clamp.

Torque Value: 11 n•m [97 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

- Install the water jacket lines if removed. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Install the engine intake manifold absolute pressure and temperature sensor. Refer to Procedure 019-099 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-099.html>)

- Connect the wire harness attachments from the air intake connection.
- Connect the charger air cooler connection. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/99/99-010-027.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for air leaks.

Last Modified: 25-Nov-2024
